Problema 1

DATI: $R_1 = 300k\Omega$, $R_2 = 200k\Omega$, $R_4 = 16\ k\Omega$, $R_5 = 1.5k\Omega$, $R_i = 40k\Omega$, $R_L = 1.2k\Omega$, $V_{DD} = 5V$

Parametrati dei MOS: M_1: $k_{n1} = 2mA/V^2$, $V_{TN1} = 0.5V$, $\lambda_{n1} = 0$

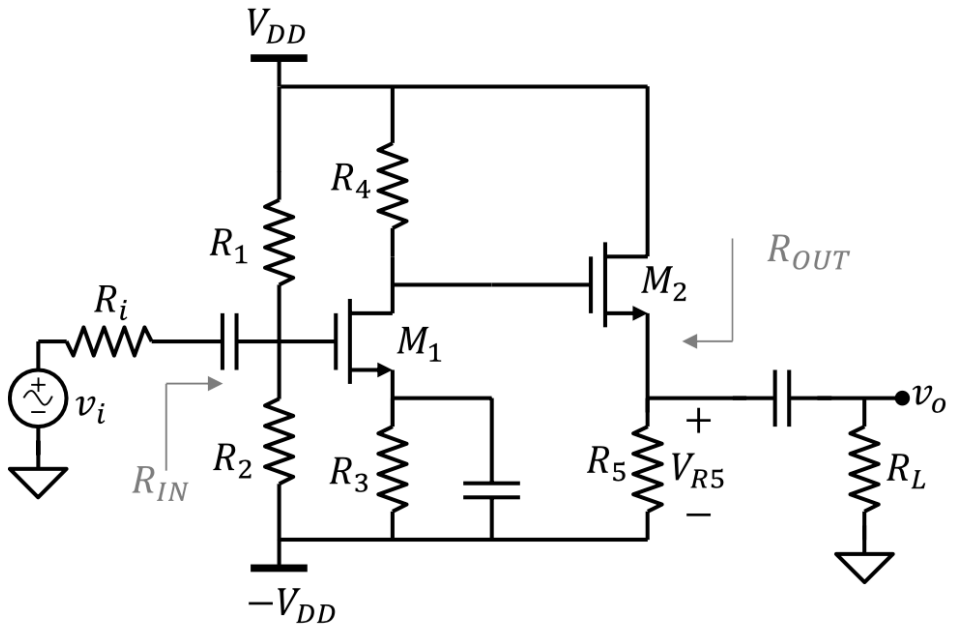
M_2: $k_{n2} = 6mA/V^2$, $V_{TN2} = 0.5V$, $\lambda_{n2} = 0$

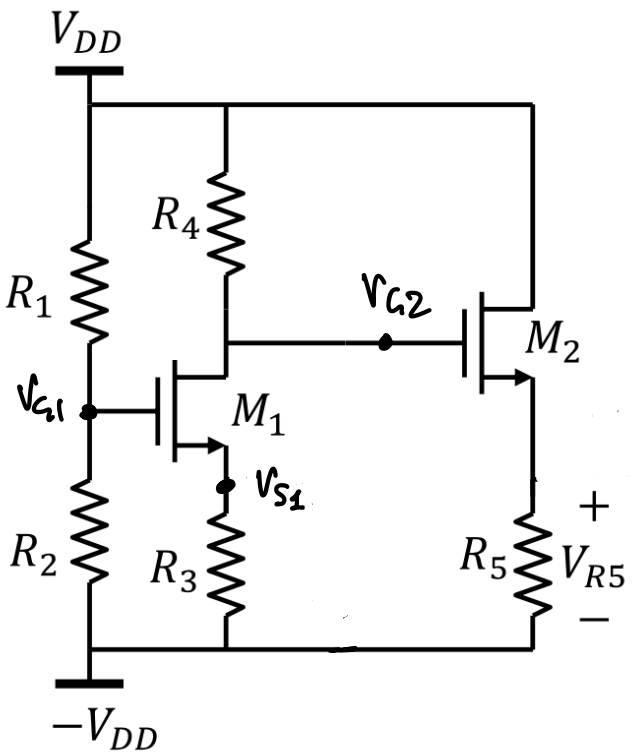
Dato il circuito in figura, calcolare:

1. Il valore delle resistenze R_3 sapendo che la corrente attraverso il MOSFET M_1 è $I_{DS1} = 0.25mA$.
2. La polarizzazione di tutti i transistor identificando la regione di funzionamento e i valori delle tensioni V_{GS} e V_{DS} e della corrente I_{DS}.
3. Disegnare il modello ai piccoli segnali e calcolare la transconduttanza g_{m1} e g_{m2} di M_1 e M_2.

Dal modello ai piccoli segnali calcolare:

4. La resistenza di ingresso R_{IN}
5. La resistenza di uscita R_{OUT}
6. Il guadagno di tensione da v_i a v_o.





1) POLARIZZAZIONE

$$V_{D1} = V_{G2} = V_{DD} - R_4 I_{D1} = 1V$$

$$V_{G1} = V_{DD} \cdot \frac{R_2}{R_1+R_2} - V_{DD} \frac{R_1}{R_1+R_2} =$$

$$= +5V \cdot \frac{2}{5} - 5V \cdot \frac{3}{5} = -1V$$

$$V_{GS1} = V_{TN1} + \sqrt{\frac{2I_{D1}}{k_{n1}}}$$

$$= 0{,}5V + 0{,}5V = 1V$$

$$\Rightarrow \quad V_{S1} = V_{G1} - V_{GS1} = -2V$$

$$V_{R3} = V_{S1} - (-V_{DD}) = 3V$$

$$\Rightarrow \quad R_3 = \frac{V_{S3}}{I_{D1}} = \frac{3V}{0{,}25mA} = 12\,K\Omega$$

$$V_{DS1} = V_{D1} - V_{S1} = 1V - (-2V) = 3V$$

$$V_{DS1} > V_{GS1} - V_{TN1} = 0{,}5V$$

OK M_1 SATURAZIONE

$$\begin{cases} V_{G2} = V_{GS2} + R_S I_{D2} - V_{DD} \\ I_{D2} = \frac{K_{n2}}{2}\left(V_{GS2} - V_{TN2}\right)^2 \end{cases} \qquad I_{D2} = \frac{V_{G2} + V_{DD}}{R_S} - \frac{V_{GS2}}{R_S}$$

$$\Rightarrow \quad \frac{2(V_{G2} + V_{DD})}{R_S K_{n2}} - \frac{2V_{GS2}}{R_S K_{n2}} = V_{GS2}^2 + V_{TN2}^2 - 2V_{GS2}V_{TN2}$$

$$V_{GS2}^2 + V_{GS2}\left(\frac{2}{R_S K_{n2}} - 2V_{TN2}\right) + V_{TN2}^2 - \frac{2(V_{G2} + V_{DD})}{R_S K_{n2}} = 0$$

$a = 1$ $\qquad b = -0{,}778$ $\qquad c = -1{,}083$

$$K_{n2} R_S = 6mA/V^2 \cdot 1{,}5\,K\Omega = 9\,V^{-1}$$

$$V_{GS2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = 1{,}5V$$

$$I_{D2} = \frac{K_{n2}}{2}\left(V_{GS2} - V_{TN2}\right)^2 = 3\,mA$$

$$V_{DS2} = V_{DD} - (-V_{DD} + V_{RS}) = 10V - 4{,}5V = 5{,}5V$$

$$V_{DS2} > V_{GS2} - V_{TN2} = 1V$$

OK M_2 SATURAZIONE

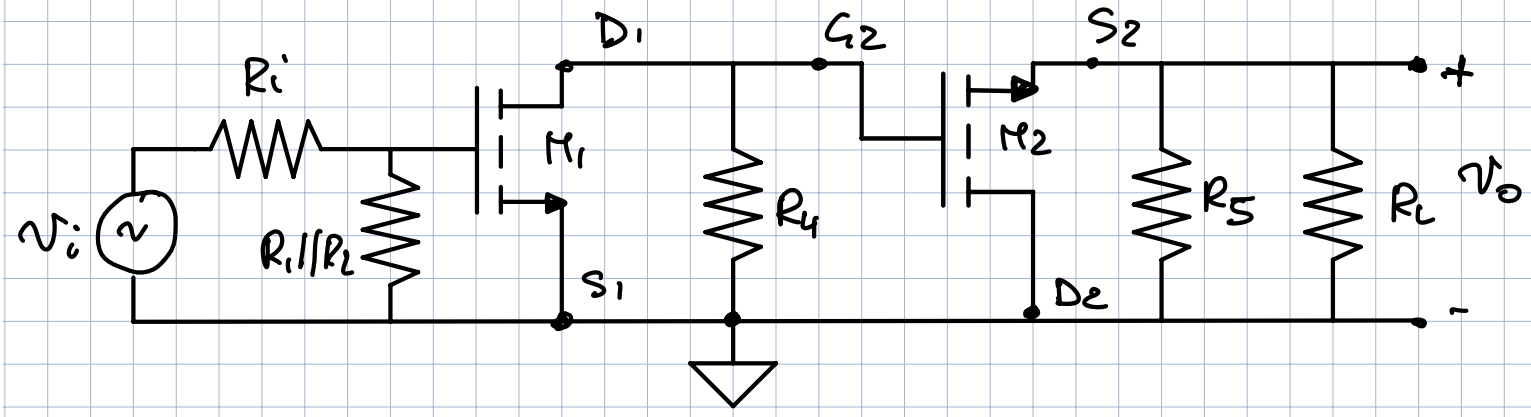
$$\left[\begin{array}{l} M_1: \; I_{D1} = 0{,}25\,mA, \; V_{DS1} = 3V, \; V_{GS1} = 1V \\ M_2: \; I_{D2} = 3\,mA, \; V_{DS2} = 5{,}5V, \; V_{GS2} = 1{,}5V \end{array}\right]$$

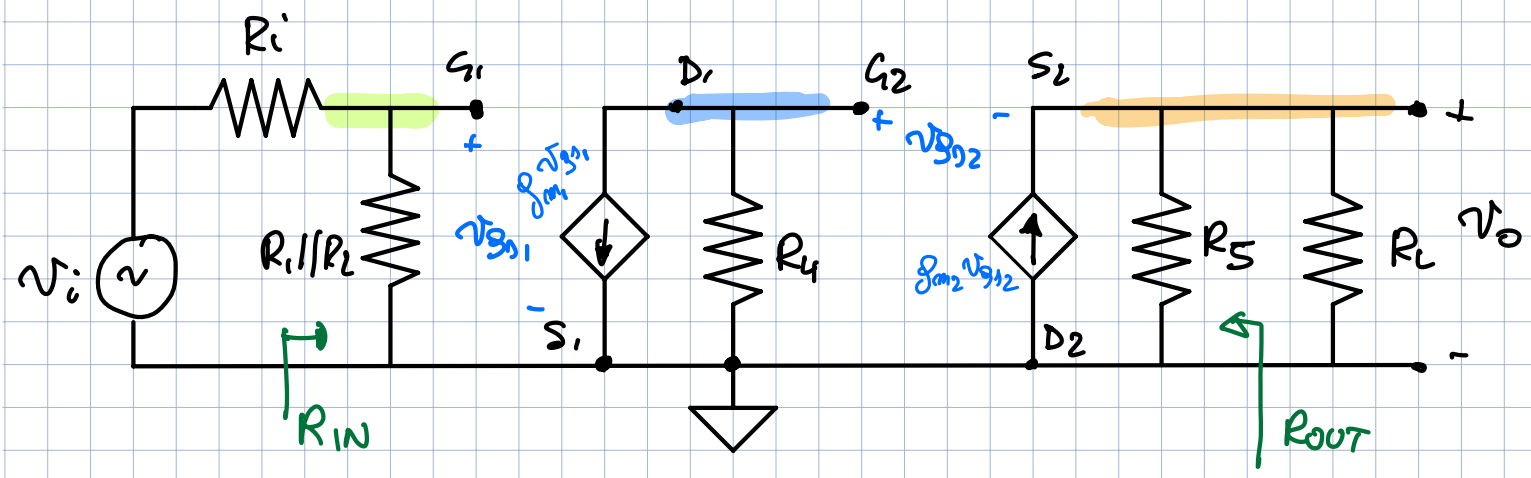
PARAMETRI AL PICCOLO SEGNALE

$$g_{m1} = \frac{2I_{D1}}{V_{GS1} - V_{TN1}} = \frac{0{,}5\,mA}{0{,}5V} = 1\,mS \qquad r_{o1} = \infty$$

$$g_{m2} = \frac{2I_{D2}}{V_{GS2} + V_{TN2}} = \frac{6\,mA}{1V} = 6\,mS \qquad r_{o2} = \infty$$

ANALISI AL PICCOLO SEGNALE





$$R_{IN} = R_1 // R_2 = 120\,K\Omega$$

$$R_{OUT} = R_5 // \frac{1}{g_{m2}} = 150\,\Omega$$

$$R_5 // R_L = 0{,}667\,K\Omega$$

$$A_v = \frac{v_o}{v_i} = \frac{v_o}{v_{g2}} \cdot \frac{v_{g2}}{v_{g1}} \cdot \frac{v_{g1}}{v_i}$$

$$\frac{v_{g1}}{v_i} = \frac{R_W}{R_W + R_i} = \frac{120}{160} = 0{,}75$$

$$\frac{v_{g2}}{v_{g1}} = g_{m1} R_d = 1\,mS \cdot 16\,k\Omega = 16$$

$$\frac{v_o}{v_{g2}} = \frac{g_{m2} R_S // R_L}{1 + g_{m2} R_S // R_L} = \frac{6\,mS \times 0{,}667}{1 + 6\,mS \times 0{,}667} = \frac{4}{5} = 0{,}8$$

$$\Rightarrow \boxed{A_v = 9{,}6}$$

non richiesto ma lo calcoliamo

$$A_i = \frac{i_o}{i_i} = \frac{\frac{v_o}{R_L}}{\frac{v_i}{R_i + R_W}} = A_v \cdot \frac{R_i + R_W}{R_L} = \frac{40\,k + 120\,k\Omega}{1{,}2\,k\Omega} A_v$$

$$= \frac{160}{1{,}2} \cdot A_v$$

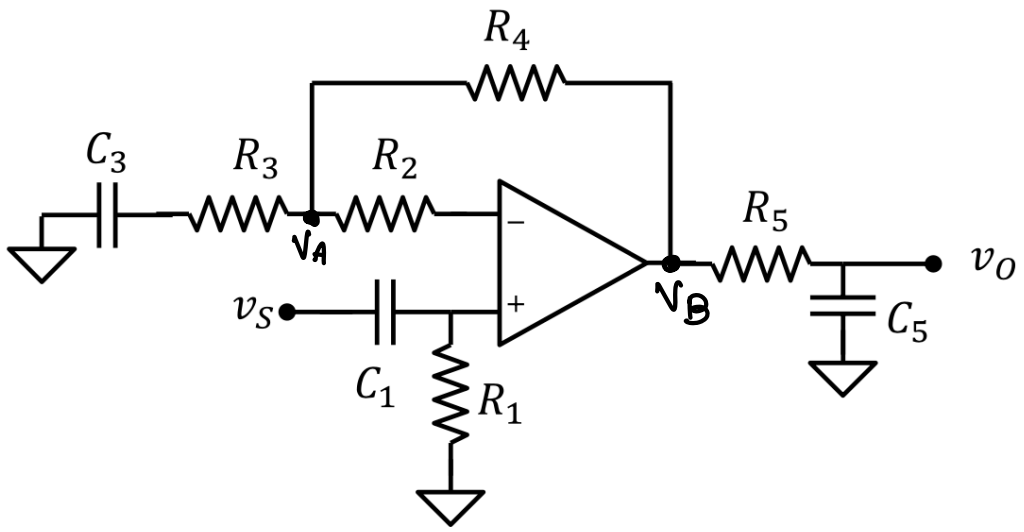
$$= \underline{\underline{1280}}$$

Problema 2

DATI: $R_1 = 100k\Omega$, $C_1 = 1\mu F$, $R_2 = 90k\Omega$, $R_3 = 10k\Omega$, $C_3 = 1nF$, $R_4 = 990k\Omega$, $R_5 = 1k\Omega$, $C_5 = 1nF$

Dato il filtro in figura realizzato con un amplificatore operazionale ideale:

1. Trovare la funzione di trasferimento del filtro $W(\omega) = v_O / v_S$.
2. Tracciare il diagramma asintotico di Bode del modulo e della fase
3. Stimare modulo e fase della funzione di trasferimento dal <u>diagramma asintotico di bode</u> per $\omega=0$ e $\omega=10^5$rad/s



N.B. SU R_2 NON PASSA CORRENTE $\Rightarrow V_{R_2} = 0$

$$\Rightarrow \boxed{v_- = V_A}$$

$$v_+ = v_S \cdot \frac{R_1}{R_1 + \frac{1}{sC_1}} = \frac{sC_1R_1}{1+sC_1R_1} v_S$$

$$Z_3 = \frac{1}{sC_3} + R_3 = \frac{1+sC_3R_3}{sC_3}$$

$$V_B = v_+ \left(1 + \frac{R_4}{Z_3}\right) = \frac{sC_1R_1}{1+sC_1R_1} \cdot \left(1 + \frac{sC_3R_4}{1+sC_3R_3}\right) v_S$$

$$= \frac{sC_1R_1}{1+sC_1R_1} \cdot \left[\frac{1+sC_3(R_3+R_4)}{1+sC_3R_3}\right] v_S$$

$$v_O = V_B \cdot \frac{\frac{1}{sC_5}}{\frac{1}{sC_5} + R_5} = V_B \cdot \frac{1}{1+sC_5R_5}$$

$$\Rightarrow W(s) = \frac{v_o}{v_s} = \frac{sC_1R_1\left[1+sC_3(R_3+R_4)\right]}{(1+sC_1R_1)(1+sC_3R_3)(1+sC_5R_5)}$$

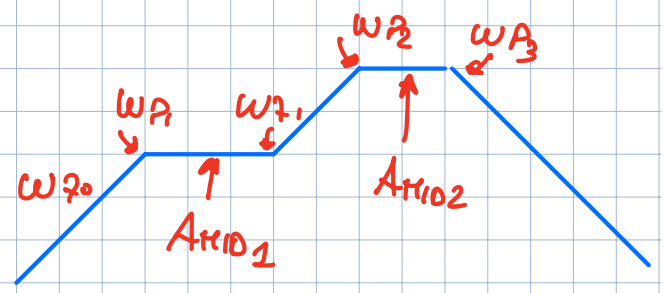
$$\omega_{z0} = \frac{1}{C_1R_1} = 10 \text{ rad/sec}$$

$$\omega_{z1} = \frac{1}{C_3(R_3+R_4)} = 10^3 \text{ rad/sec}$$

$$\omega_{p1} = \frac{1}{C_1R_1} = 10 \text{ rad/sec}$$

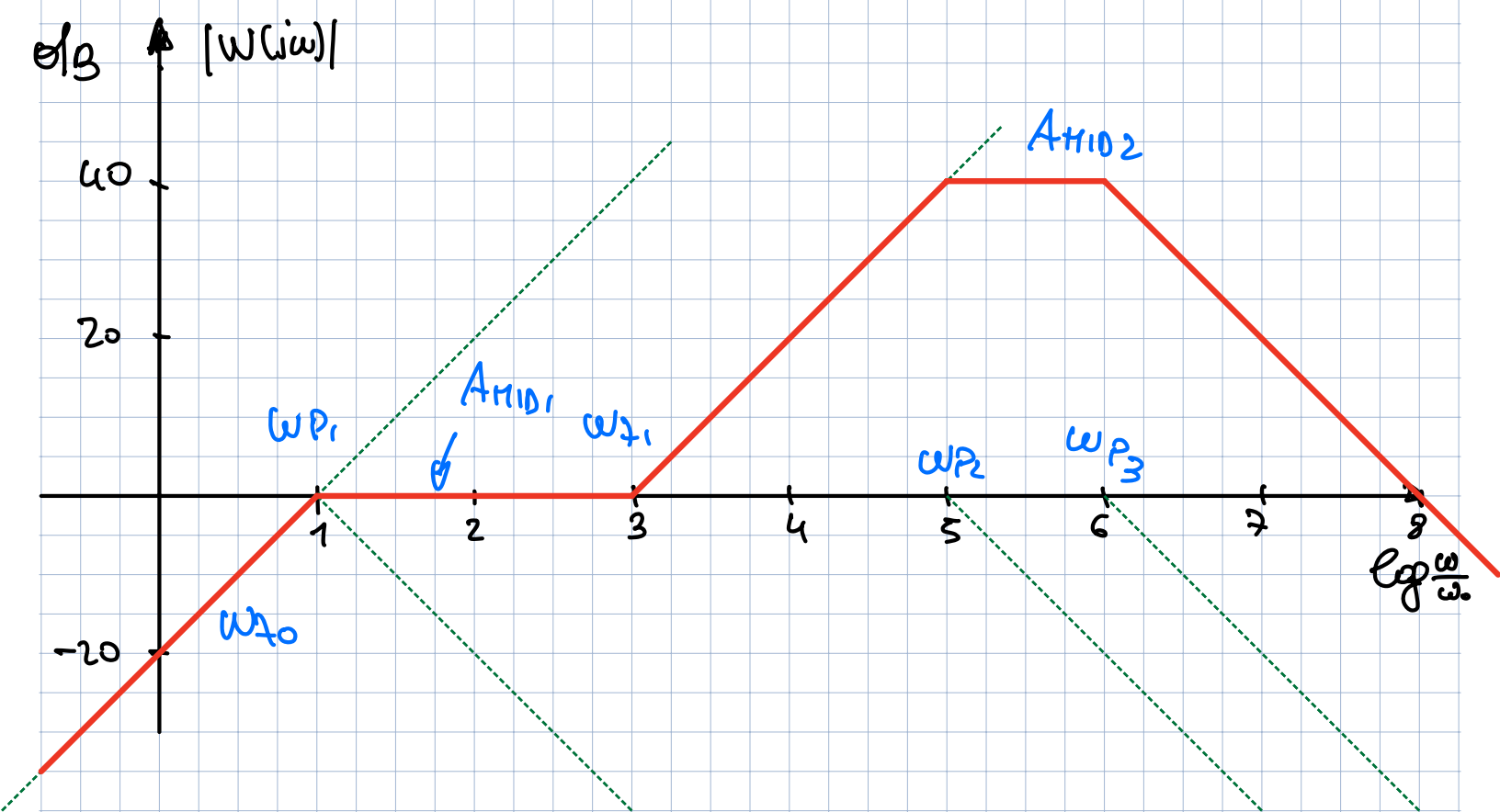
$$\omega_{p2} = \frac{1}{C_3R_3} = 10^5 \text{ rad/sec}$$

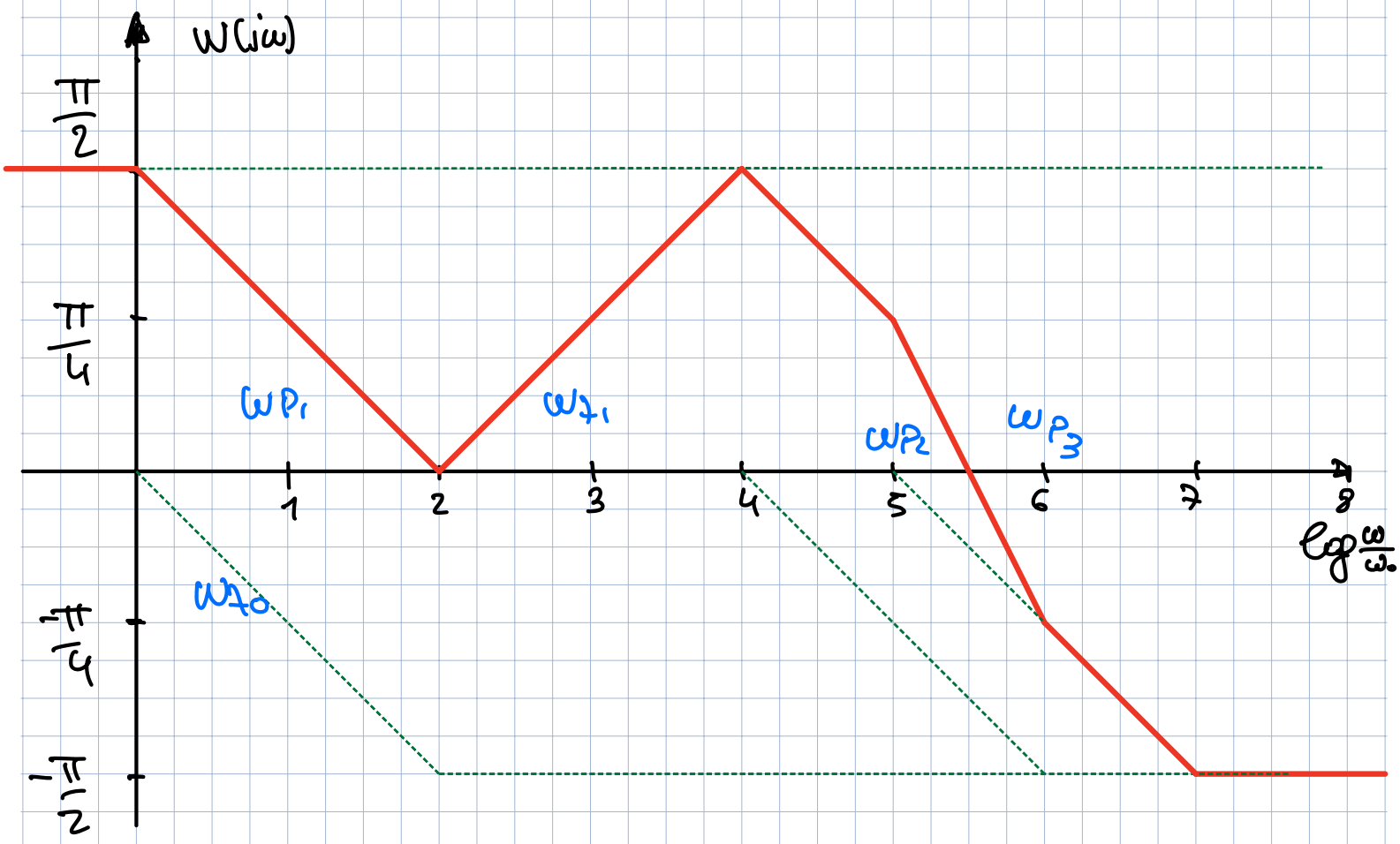
$$\omega_{p3} = \frac{1}{C_5R_5} = 10^6 \text{ rad/sec}$$



$$A_{MID1} = \frac{C_1R_1}{C_1R_1} = 1 = 0\,dB$$

$$A_{MID2} = \frac{C_1R_1C_3(R_3+R_4)}{C_1R_1 \cdot C_3R_3} = \frac{R_3+R_4}{R_3} = 100 = 40\,dB$$





$$\omega = 0 \Rightarrow |W(j\omega)| = 0$$

$$\angle W(j\omega) = \frac{\pi}{2}$$

$$\omega = 10^5 \qquad |W(j\omega)| = \frac{100}{\sqrt{2}}$$

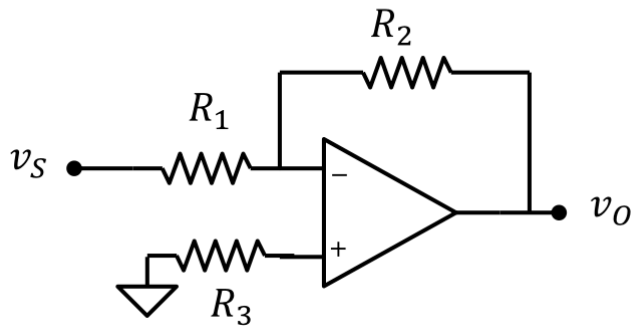
$$\angle W(j\omega) = \frac{\pi}{4}$$

Problema 3

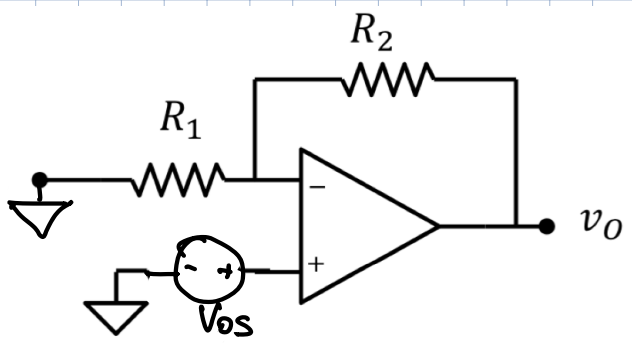
DATI: $R_1 = 30k\Omega$, $R_2 = 150k\Omega$

Sia dato il circuito in figura realizzato con un amplificatore operazionale reale con una tensione di offset $V_{OS} = 2mV$ e correnti di bias $I_{BP} = 100nA$ e $I_{BN} = 80nA$.

1. Assumendo $R_3 = 100\ k\Omega$ e $v_S = 20mV$, calcolare la tensione v_O.
2. Quanto deve valere R_3 per annullare l'effetto delle correnti di bias?
3. Esiste un valore di R_3 che permette di annullare sia le correnti di bias che la tensione di offset? Se si calcolarlo.

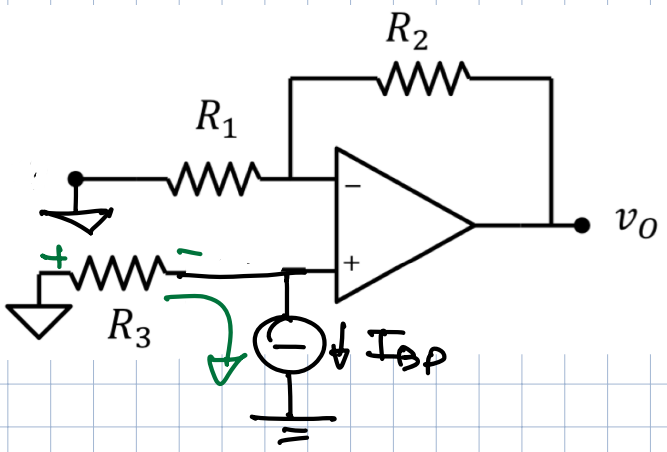


1a) EFFETTO V_{OS}



$$v_{O1} = V_{OS}\left(1+\frac{R_2}{R_1}\right) = 2mV\left(1+\frac{150}{30}\right) = 12mV$$

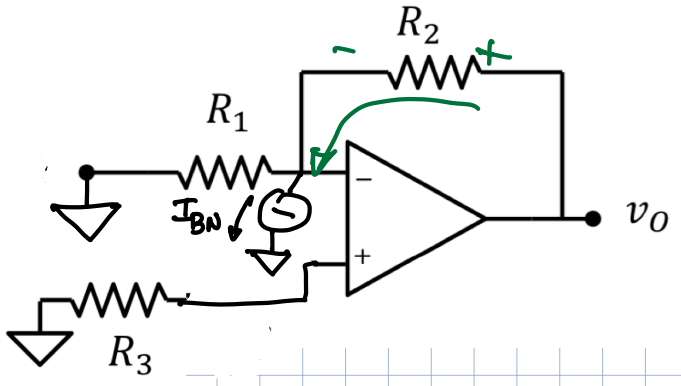
1b) EFFETTO I_{BP}



$$v_+ = -I_{BP}R_3 = 100nA \cdot 100k\Omega = 10^{-7}A \cdot 10^5\Omega = 10^{-2}V = 10mV$$

$$v_{O2} = v_+\left(1+\frac{R_2}{R_1}\right) = -60mV$$

1c) EFFETTO I_{BN}



$I_{R_1} = 0 \quad v_+ = v_- = 0$

$$v_{O3} = I_{BN} \cdot R_2 = 80nA \cdot 150k\Omega = 12mV$$

$\Rightarrow$ EFFETTO DI $V_{OS} + I_{BN} + I_{BP}$:

$$V_{o1} + V_{o2} + V_{o3} = -36\ mV$$

EFFETTO DELLA SOLA $V_S = 20\ mV$

$$V_{o4} = -V_S\left(\frac{R_2}{R_1}\right) = -5V_S = -100\ mV$$

EFFETTO COMPLESSIVO:

$$\Rightarrow \quad V_o = V_{o1} + V_{o2} + V_{o3} + V_{o4} = -136\ mV$$

2) PER ANNULLARE I_{BIAS}?

$$I_{BP} \cdot R_3\left(1 + \frac{R_2}{R_1}\right) = I_{BN} R_2$$

$$R_3 = \frac{I_{BN}}{I_{BP}} R_2 \left(\frac{R_1}{R_1 + R_2}\right) = \frac{I_{BN}}{I_{BP}} R_1 \| R_2 = \boxed{20\ K\Omega}$$

CON $V_S = 0$

$$V_o = 12\ mV \quad (\text{SOLO effetto } V_{OS})$$

CON $V_S = 20\ mV$

$$V_o = V_{o1} + V_{o4} = -88\ mV$$

3) POSSO ANNULLARE V_{OS} e I_B? SÌ:

$$V_{OS}\left(1 + \frac{R_2}{R_1}\right) - I_{BP} R_3\left(1 + \frac{R_2}{R_1}\right) + I_{BN} R_2 = 0$$

$$R_3 I_{BP}\left(1 + \frac{R_2}{R_1}\right) = V_{OS}\left(1 + \frac{R_2}{R_1}\right) + I_{BN} R_2$$

$$R_3 = \frac{V_{OS}}{I_{BP}} + \frac{I_{BN}}{I_{BP}} R_1 \| R_2$$

$$\Rightarrow \boxed{R_3 = 40\ K\Omega} \qquad \frac{V_{OS}}{I_{BP}} = \frac{2\ mV}{10^{-7} A} = 20\ K\Omega$$